

Comparison of Erasure-Aware Error Correction Decoding for Fault Tolerant Quantum Computing

University of California, Berkeley
C191A Introduction to Quantum Computing I
Final Project. Fall 2025

Leigh Zaman Arin Sen
`leighzaman@berkeley.edu` `arin.sen@berkeley.edu`

Ilyes Tata
`ilyes.tata@berkeley.edu`

Last revised:
November 22, 2025

Abstract

TODO

The abstract should summarize our main points: purpose, methods, results, and conclusions.
Keep it short.

Contents

1	Introduction	3
2	Background & Related Work	3
2.1	Noise Models	3
2.2	Erasure Qubits	4
2.2.1	Advantages, Challenges of Erasure Errors	4
2.2.2	Formalism	5
3	Methods	5
4	Metrics	6
5	Results	6
6	Discussion	6
7	Conclusion	6
8	References	7
A	Appendix: Syndrome of Loss Errors	9
B	Appendix: Additional Figures and Tables	9
C	Glossary	10

1 Introduction

Motivation & research question

Motivation: Fault-tolerate quantum computation at scale remains a formidable challenge, but we’re excited about these recent major advancements:

Decoding with erasure information dramatically increases the surface code error threshold from 0.937% to 4.15%, with an estimated 98% of errors convertible into erasures. [1]

Machine-learned error decoding for quantum processors (surface code) by Bausch et al uses a transformer-based neural network to decode error data with realistic noise models, including qubit cross-talk and state leakage.

"Requires a system capable of dissipatively removing physical errors while simultaneously performing arbitrary coherent manipulation of the encoded logical qubits." [2]

Sub-questions & hypotheses

2 Background & Related Work

2.1 Noise Models

PAULI NOISE. The simplest model used to evaluate Quantum Error Correction (QEC) assumes all physical qubit errors can be represented by uniformly random Pauli operators applied during circuit simulation. In our simulations we implement a [depolarizing error channel](#) to represent [Pauli noise](#), which has the following effect on some quantum state ρ with p probability of becoming a maximally-mixed state and $(1 - p)$ probability of remaining unchanged:

$$E_{\text{pauli}}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad [\text{Depolarizing error channel}] \quad (1)$$

Unfortunately, error models based only on [Pauli noise](#) are not robust enough to describe the real-world noise that must be corrected to realize fault-tolerant quantum computation. [CITATION NEEDED]

LEAKAGE. A [leakage error](#) occurs when a qubit “leaks” out of the computational subspace (described by a two-dimensional Hilbert space) into a state that can only be described with a higher-order Hilbert space. If leakage goes undetected, the error compounds over time as it spreads through a multi-qubit interactions. Leakage can be stochastically modeled with a 3-level system (qutrit) with orthonormal basis: $\{ |0\rangle, |1\rangle, |2\rangle \}$. Leakage events are assumed to be discrete and occur independently with probability p_{leak} at each gate operation acting on the qutrit [3]. The leakage error $E_{E_{\uparrow}}$ on the j th qutrit in the system can then be understood as:

$$E_{\text{leakage}}(\rho_j) = (1 - p_{\text{leak}})\rho_j + p_{\text{leak}} |2\rangle \langle 2| \quad [\text{Leakage error channel [3]}] \quad (2)$$

Leakage-reduction units (LRUs) were proposed as a way to convert leakage errors into Pauli-type errors, based on a one-bit teleportation protocol that effectively moves the logical state from the physical qubit in the leaked state to a freshly-prepared physical qubit [3].

All species of qubit experience leakage errors, but it’s important to note the underlying *physical* processes look and behave very differently despite sharing a common mathematical description. Neutral atoms (Rydberg atoms)

Transmon leakage TODO [4]

Neutral atom / Rydberg leakage. Neutral atom entangling gates exploit Rydberg blockade, which is a quantum phenomenon that occurs when an atom is excited to a particular Rydberg

energy level, which prevents nearby atoms from being excited to the same state. The excitation of the single valence electron in alkali metals like Cs, Rb results in a large effective atomic radius [CITATION NEEDED, but I think it's on the order of 25E5], creating a large electric dipole moment.

(one or more electrons in the case of alkaline earth metals or lanthanides like Yb) TODO [5]

ERASURE. An **erasure error** occurs when a qubit is longer detectable by the measurement device, so the quantum state information encoded by the qubit is effectively “lost” or “erased”. Mathematically, we extend the qubit Hilbert space with some $|e\rangle$ that is orthogonal to the qubit state $\{|0\rangle, |1\rangle\}$. The erasure channel can be defined as:

$$E_{\text{erase}}(\rho_j) = (1 - p_{\text{erase}})\rho_j + p_{\text{erase}} |e\rangle \langle e| \quad [\text{Erasure error channel [3]}] \quad (3)$$

Photon loss is the dominant form of error in photonic qubits [CITATION NEEDED].

Leakage-detection units (LDUs) were developed to address leakage errors in superconducting qubits, by periodically coupling a data qubit to an ancilla that flags when the data qubit leaves the computational subspace. The flagged data qubit can be reset/replaced during subsequent error-correction cycles, effectively treating the flagged qubit as an erasure error.

Neutral atoms / Rydberg arrays, each atom is arranged into a known position using optical tweezers. Although the lattice/trap is situated inside a vacuum chamber, some nonzero amount of background gas collisions or thermal energy transfer from the environment can eject an atom from its lattice site.

CORRELATED ERRORS. TODO The most important thing to mention is that real-world errors are often correlated, so fault-tolerant QC requires correction of correlated noise. [6]

Cross-talk might also be worth mentioning for completeness.

2.2 Erasure Qubits

A qubit engineered so the dominant source of errors is from erasures.

A bias towards erasure-type errors occurs natively in some types of qubits (e.g. photons) or can be engineered by converting leakage errors to erasures (e.g. with LDUs in superconducting qubits).

2.2.1 Advantages, Challenges of Erasure Errors

ADVANTAGES. From: "The benefits and costs of quantum error correction with erasure qubits" (QIP 2025)

1. Can theoretically correct more erasure errors than unknown Pauli errors. We can correct $d-1$ erasure errors compared to $\frac{d-1}{2}$ Pauli errors. As a result, we expect logical error rate P_L on the order of p^d where p is the physical qubit error rate, compared to $p^{d/2}$ for unknown Pauli errors. **gu_2023**
2. Higher thresholds: surface code 50% erasure noise vs. 19% depolarizing noise.
3. Dominant noise source can be engineered

TODO Figure: d=5 surface code with some unknown Pauli error spanning 3 qubits, with syndrome squares highlighted. If we apply the minimum-weight correction, this causes a logical error.

Figure: d=5 surface code with erasure errors, exact locations known.

CHALLENGES. Challenges from "The benefits and costs of quantum error correction with erasure qubits" (QIP 2025):

1. More physical operations.
2. Erasure detection imperfect, introducing another dimension to error models.
3. Midcircuit erasure checks can create correlated noise for neighboring qubits
4. Tools to simulate and decode erasure circuits are active area of research.

ERASURE CONVERSION IN SUPERCONDUCTING QUBITS

1. Shouzhen Gu. "Fault-tolerant quantum architectures based on erasure qubits" and "Optimizing quantum error correction protocols with erasure qubits" [7] [8]

ERASURE CONVERSION IN NEUTRAL / RYDBERG ATOMS

1. Yue Wu, Shimon Kolkowitz, Shruti Puri, Jeff D Thompson. Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays [9]
2. Gefen Baranes, Madelyn Cain, J. Pablo Bonilla Ataides, Dolev Bluvstein, Josiah Sinclair, Vladan Vuletic, Hengyun Zhou, Mikhail D. Lukin . Leveraging Atom Loss Errors in Fault Tolerant Quantum Algorithms [10]

2.2.2 Formalism

define stabilizer circuits with erasure channel: [8]
compare w/ surface code having only pauli error channel

3 Methods

SIMULATING ERASURE CIRCUITS Describe Stim circuits. Link to Stim circuits / appendix figures of selected timeline-svg operations. [11]

DECODING TODO

Describe the decoding problem for Pauli error stabilizer circuits in the surface code. Explain how surface code error syndromes can be framed as a graph traversal problem with detection events as nodes and errors as the weighted edges. The minimum weight perfect matching (MWPM) algorithm can find the most probable set of errors that occurred. Explain why MWPM works for uniformly random Pauli errors, but why it experiences a sharp drop-off in performance for biased (non-identical) noise models.

Describe the decoding problem when 100 of noise channels can be converted to erasure errors (conversion_ratio=1 in our codebase). Describe how the Hypergraph Minimum-Weight Parity Factor (MWPF). [12]

4 Metrics

5 Results

6 Discussion

7 Conclusion

ACKNOWLEDGEMENTS Shimon Kolkowitz, Harry Levine, Thiago Bergamaschi

8 References

- [1] Y. Wu, S. Kolkowitz, S. Puri, and J. D. Thompson, “Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays,” en, *Nature Communications*, vol. 13, no. 1, p. 4657, Aug. 2022, Publisher: Nature Publishing Group, ISSN: 2041-1723. DOI: [10.1038/s41467-022-32094-6](https://doi.org/10.1038/s41467-022-32094-6). Accessed: Nov. 14, 2025. [Online]. Available: <https://www.nature.com/articles/s41467-022-32094-6>.
- [2] D. Bluvstein *et al.*, *Architectural mechanisms of a universal fault-tolerant quantum computer*, arXiv:2506.20661 [quant-ph], Jun. 2025. DOI: [10.48550/arXiv.2506.20661](https://doi.org/10.48550/arXiv.2506.20661). Accessed: Nov. 5, 2025. [Online]. Available: <http://arxiv.org/abs/2506.20661>.
- [3] M. Suchara, A. W. Cross, and J. M. Gambetta, *Leakage Suppression in the Toric Code*, arXiv:1410.8562 [quant-ph], Oct. 2014. DOI: [10.48550/arXiv.1410.8562](https://doi.org/10.48550/arXiv.1410.8562). Accessed: Nov. 22, 2025. [Online]. Available: <http://arxiv.org/abs/1410.8562>.
- [4] Z. Chen *et al.*, “Measuring and Suppressing Quantum State Leakage in a Superconducting Qubit,” *Physical Review Letters*, vol. 116, no. 2, p. 020 501, Jan. 2016, arXiv:1509.05470 [quant-ph], ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.116.020501](https://doi.org/10.1103/PhysRevLett.116.020501). Accessed: Nov. 22, 2025. [Online]. Available: <http://arxiv.org/abs/1509.05470>.
- [5] M. Morgado and S. Whitlock, “Quantum simulation and computing with Rydberg-interacting qubits,” en, *AVS Quantum Science*, vol. 3, no. 2, p. 023 501, Jun. 2021, ISSN: 2639-0213. DOI: [10.1116/5.0036562](https://doi.org/10.1116/5.0036562). Accessed: Oct. 17, 2025. [Online]. Available: <https://pubs.aip.org/aqs/article/3/2/023501/997323/Quantum-simulation-and-computing-with-Rydberg>.
- [6] C. D. Wilen *et al.*, “Correlated charge noise and relaxation errors in superconducting qubits,” en, *Nature*, vol. 594, no. 7863, pp. 369–373, Jun. 2021, Publisher: Nature Publishing Group, ISSN: 1476-4687. DOI: [10.1038/s41586-021-03557-5](https://doi.org/10.1038/s41586-021-03557-5). Accessed: Nov. 22, 2025. [Online]. Available: <https://www.nature.com/articles/s41586-021-03557-5>.
- [7] S. Gu, Y. Vaknin, A. Retzker, and A. Kubica, *Optimizing quantum error correction protocols with erasure qubits*, en, arXiv:2408.00829 [quant-ph], Aug. 2024. DOI: [10.48550/arXiv.2408.00829](https://doi.org/10.48550/arXiv.2408.00829). Accessed: Nov. 13, 2025. [Online]. Available: <http://arxiv.org/abs/2408.00829>.
- [8] S. Gu, A. Retzker, and A. Kubica, “Fault-tolerant quantum architectures based on erasure qubits,” *Physical Review Research*, vol. 7, no. 1, p. 013 249, Mar. 2025, arXiv:2312.14060 [quant-ph], ISSN: 2643-1564. DOI: [10.1103/PhysRevResearch.7.013249](https://doi.org/10.1103/PhysRevResearch.7.013249). Accessed: Nov. 6, 2025. [Online]. Available: <http://arxiv.org/abs/2312.14060>.
- [9] J. Thompson, *Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays*, en, Jun. 2022. DOI: [10.7910/DVN/H9LV4H](https://doi.org/10.7910/DVN/H9LV4H). Accessed: Oct. 18, 2025. [Online]. Available: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/H9LV4H>.
- [10] G. Baranes *et al.*, *Leveraging Atom Loss Errors in Fault Tolerant Quantum Algorithms*, arXiv:2502.20558 [quant-ph] version: 1, Feb. 2025. DOI: [10.48550/arXiv.2502.20558](https://doi.org/10.48550/arXiv.2502.20558). Accessed: Nov. 1, 2025. [Online]. Available: <http://arxiv.org/abs/2502.20558>.
- [11] C. Gidney, *Stim: A fast stabilizer circuit simulator*, Pages: 497 Publication Title: Quantum Volume: 5 original-date: 2021-02-19T15:55:47Z, Jul. 2021. Accessed: Oct. 19, 2025. [Online]. Available: <https://doi.org/10.22331/q-2021-07-06-497>.

- [12] Y. Wu, B. Li, K. Chang, S. Puri, and L. Zhong, *Minimum-Weight Parity Factor Decoder for Quantum Error Correction*, arXiv:2508.04969 [quant-ph], Oct. 2025. DOI: [10.48550/arXiv.2508.04969](https://arxiv.org/abs/2508.04969). Accessed: Nov. 18, 2025. [Online]. Available: <http://arxiv.org/abs/2508.04969>.

A Appendix: Syndrome of Loss Errors

Describe how a loss is heralded during state readout. Each potential loss location is described by a distinct error probability, described by a unique configuration of the error syndroms. [EXAMPLE OF ERROR SYNDROME]

B Appendix: Additional Figures and Tables

C Glossary

depolarizing error channel [1](#)

Pauli noise A widely-used error model that characterizes noise as a random combination of Pauli-X, Pauli-Y, and Pauli-Z operations. Many early benchmarks and error correction algorithm thresholds assume an independent and identically distributed Pauli noise model, but real-world error models are more complex. [1](#)